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Analyzing Multiple Multivariate Time Series Data Using Multilevel Dynamic Factor Models

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Multivariate time series data offer researchers opportunities to study dynamics of various systems in social and behavioral sciences. Dynamic factor model (DFM), as an idiographic approach for studying intraindividual variability and dynamics, has typically been applied to time series data obtained from a single unit. When multivariate time series data are collected from multiple units, how to synchronize dynamical information becomes a silent issue. To address this issue, the current study presented a multilevel dynamic factor model (MDFM) that analyzes multiple multivariate time series in multilevel SEM frameworks. MDFM not only disentangles within- and between-person variability but also models dynamics of the intraindividual processes. To illustrate the uses of MDFMs, we applied lag0, lag1, and lag2 MDFMs to empirical data on affect collected from 205 dating couples who had at least 50 consecutive days of observations. We also considered a model extension where the dynamical coefficients were allowed to be randomly varying in the population. The empirical analysis yielded interesting findings regarding affect regulation and coregulation within couples, demonstrating promising uses of MDFMs in analyzing multiple multivariate time series. In the end, we discussed a number of methodological issues in the applications of MDFMs and pointed out possible directions for future research.

Time series data refer to observations that are collected repeatedly and intensively for phenomena changing over time. Some examples include daily recordings of temperature, daily ratings of a person's emotion, and minutely recordings of physiological activities. The primary goal of analyzing time series data is to discover dynamics that govern the systems unfolding over time. In social and behavioral sciences, researchers are interested in investigating dynamics of human behaviors, complex systems that typically require collecting multivariate time series—intensive, repeated observations on a number of variables.

In recent years, statistical techniques for analyzing multivariate time series data have been developing rapidly. For example, as a continuous-time modeling method, latent differential equation models were developed to model intrinsic dynamics of latent psychological processes through utilizing dampened linear oscillator models (e.g., Boker, Neale, & Rausch, 2004) while constructing the measurement model across time instead of observed variables as used in standard factor-analytic models. Another example is the discrete-time state-space models, a general statistical approach for modeling intraindividual dynamics, subsuming many commonly used models as special cases (Chow, Ferrer, & Nesselroade, 2007; Chow, Ho, Hamaker & Dolan, 2010; Song & Ferrer, 2009). Due to the uses of iterative estimation procedures like Kalman filter, state-space models are typically implemented using special computer software (e.g., mkfm6; Dolan, 2005) that is not readily used by applied researchers. The third example, the focal model in this article, is dynamic factor models that were developed to model dynamics of latent processes by combining factor-analytic and time series techniques. Although dynamic factor models can be represented as a special case of state-space model and therefore estimated using the state-space approach (see Chow et al., 2010), our

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focus here is to represent and estimate dynamic factor models as structural equation models, which would be easily understood and adopted by applied researchers.

Dynamic factor models (DFMs; McArdle, 1982; Molenaar, 1985) have been widely used in behavioral sciences to study dynamical mechanisms underlying a variety of behavioral systems (e.g., Chow, Nesselroade, Shifren, & McArdle, 2004; Chow, Zu, Shifren, & Zhang, 2011; Ferrer & Nesselroade, 2003; Molenaar, 1985, 1987; Nesselroade & Molenaar, 1999; Ram, Brose, & Molenaar, 2013; Shifren, Wood, Hooker, & Molenaar, 1997; Wood & Brown, 1994). A typical DFM consists of two equations.[1] The first one is a factor-analytic model used to take into account measurement errors and extract common factors. The second equation is a time series model used to model dynamics (i.e., autoregressive correlations and cross-correlations) of the factor scores over time. Mathematical expressions of a first-order DFM (lag1 DFM) are as follows:

$$\mathbf{y_t} = \boldsymbol{\Lambda}\boldsymbol{\eta}_\mathbf{t} + \boldsymbol{\varepsilon}_\mathbf{t}, \quad (1)$$

$$\boldsymbol{\eta}_\mathbf{t} = \mathbf{B}\boldsymbol{\eta}_\mathbf{t-1} + \boldsymbol{\xi}_t, \quad (2)$$

where $\mathbf{y_t}$ is a p-dimensional vector of observed time series at time t (t = 1,2, ... ,T); $\boldsymbol{\eta}_\mathbf{t}$ and $\boldsymbol{\eta}_\mathbf{t-1}$ are k-dimensional vectors of the factor scores at time t and t -1, respectively; $\boldsymbol{\Lambda}$ is the p X k factor loading matrix that is often assumed to be invariant over time; $\mathbf{B}$ is a k X k matrix containing auto- and cross-regression coefficients within and between factors; and $\boldsymbol{\varepsilon}_\mathbf{t}$ and $\boldsymbol{\xi}_t$ are vectors of error terms that are assumed to be uncorrelated. When applying DFMs, the parameter matrix $\mathbf{B}$ is of particular interest because it includes dynamical information of the system under study. One important assumption underlying DFMs is that the multivariate time series, $\mathbf{y_t}$, need to be stationary, that is, both their means and variances-covariances are constant over time.

DFMs were developed as an idiographic approach for studying intraindividual variability and its mechanisms (Nesselroade & Ford, 1985) and therefore, they have been typically applied to multivariate time series observations collected from single units of research (e.g., individual or dyad). However, as one attempt to study normative law of human behaviors, data on multiple units are typically collected in behavior science. For example, one may ask 100 participants rating their affects every day using a 20-item questionnaire for 90 consecutive days. Although possible, it seems cumbersome to apply the standard DFMs to each participant's multivariate time series observations separately. Therefore, how to synchronize dynamical information from multiple units becomes needed.

Researchers have attempted to develop efficient ways to integrate information from multiple multivariate time series in the context of DFMs.[2] For instance, Nesselroade and Molenaar (1999) developed a method to analyze multiple multivariate time series—lagged observed covariance matrices were pooled across individuals to estimate population lagged covariance matrix and then dynamic factor models were fitted to the resultant matrix. Hamaker, Nesselroade, and Molenaar (2007) proposed an integrated state-trait model by which DFMs were applied to individual multivariate time series to examine state variability and then common factor-analytic models were applied to (corrected) intraindividual means of all individuals to examine trait variability. Later, Ferrer and Widaman (2008) fitted identical dynamic factor models to each of the multiple multivariate time series and then examined the empirical distributions (e.g., sample means and variances) of parameters as a way to integrate information. Recently, Song and Ferrer (2012) presented random coefficient dynamic factor models for simultaneously modeling multiple multivariate series with which the lagged relationships among latent variables (i.e., the dynamics coefficients) were assumed to be randomly varying and potential covariates could be added to account for the variations.

Most of the current methods of analyzing multiple multivariate time series with DFM components lack the capability of handling information at the between-person level (except for Hamaker et al., 2007). In other words, the models are fitted to the within-person data only, either by pooling together individual data (e.g., Nesselroade& Molenaar, 1999) or parameter estimates (e.g., Ferrer & Widaman, 2008) or by assuming information from individual data forming a random distribution (e.g., Song & Ferrer, 2012). Therefore, although these techniques are useful in modeling within-person variability and dynamics, they are limited in exploring between-person variability and its sources that can be very important in their own right.

This study aimed to offer a novel way to analyze multiple multivariate time series by which both within-person variability/dynamics and between-person variability can be handled simultaneously. We proposed multilevel dynamic factor models (MDFMs) by integrating DFMs with multilevel modeling in the framework of multilevel structural equation models. In the following sections, we first briefly review the basics of multilevel structural equation models and then present MDFMs. We then analyze a large set of empirical data to illustrate MDFMs. Finally, the utility of the proposed model is discussed together with possible extensions and limitations.

[1]In this study, we used the dynamic factor model with the specification referred to as Process Factor Analysis model (See Browne & Nesselroade, 2005; Nesselroade, McArdle, Aggen, & Meyers, 2002, for a description of the other specification).

[2]Similar attempts have been made in other modeling contexts, such as differential structural equation models (Boker & Ghisletta, 2001; Butner, Amazeen, & Mulvey, 2005) and state-space models (Chow, Hamaker, Fujita, & Boker, 2009).

MULTILEVEL STRUCTURAL EQUATION MODELS

Multilevel structural equation model (MSEM) is an integration of multilevel regression model with structural equation model to account for measurement errors as well as clustered sampling. It is suitable for valid statistical inference when the observations are nested within clusters and some variables of interest are measured by a set of items or instruments. For example, in a study about the influence of racial discrimination on depression among ethnic minority students, the researcher may want to employ SEM to study the relationship between the two latent constructs within school and to take into account the fact that the students are recruited from multiple schools. Multilevel SEM becomes needed in this type of research context.

The basic idea behind MSEM (see Asparouhov & Muthén, 2007, 2012; Muthén, 1994) is that a variable is decomposed into a within-cluster component and between-cluster component. That is, the vector of observed variables, denoted as $\mathbf{y}_{ij}$, is decomposed into two parts,

$$\mathbf{y}_{ij} = \mathbf{y}_{1,ij} + \mathbf{y}_{2,j}, \tag{3}$$

where $\mathbf{y}_{ij}$ is a vector of observed variables for person i in cluster j and the two components, $\mathbf{y}_{1,ij}$ and $\mathbf{y}_{2,j}$, are normally distributed variables, with $\mathbf{y}_{1,ij}$ representing the within-cluster effect (i.e., individual differences) and $\mathbf{y}_{2,j}$ representing the between-cluster effect (i.e., the random intercepts). Thus, $\mathbf{y}_{1,ij}$ is defined on Level 1 and $\mathbf{y}_{2,j}$ is defined on Level 2, as indicated by the associated numerical subscripts. To help transit to multilevel dynamic factor model that is described later in this article, the MSEM presented here is a two-level multilevel factor model in which all observed variables are assumed to have within and between variations. Readers who are interested in a more general presentation of MSEM could refer to Muthén and Asparouhov (2009) and Preacher, Zyphur, and Zhang (2010), where covariates can be included at any level and observed variables could have variations at one or more levels.

The multilevel factor model is expressed as

$$\text{within-cluster}: \mathbf{y}_{1,ij} = \mathbf{\Lambda}_{1,j}\boldsymbol{\eta}_{1,ij} + \boldsymbol{\varepsilon}_{1,ij}, \tag{4}$$

$$\text{between-cluster}: \mathbf{y}_{2,j} = \boldsymbol{\nu}_2 + \mathbf{\Lambda}_2\boldsymbol{\eta}_{2,j} + \boldsymbol{\varepsilon}_{2,j}, \tag{5}$$

where the single subscript j after the commas permits the parameters to potentially vary across clusters; $\mathbf{y}_{1,ij}$ and $\mathbf{y}_{2,j}$ refer to the p-dimensional within-cluster and between-cluster components, respectively; $\boldsymbol{\eta}_{1,ij}$ (k X 1) and $\boldsymbol{\eta}_{2,j}$ (m X 1) are vectors of latent variables (m could be equal to p); $\mathbf{\Lambda}_{1,j}$(p X k) and $\mathbf{\Lambda}_2$(p X m) are factor loading matrices; $\boldsymbol{\nu}_2$is a p-dimensional intercept vector; and $\boldsymbol{\varepsilon}_{1,ij}$ and $\boldsymbol{\varepsilon}_{2,j}$ are both p X 1 vectors of residual terms, normally distributed with zero means and covariance matrix $\mathbf{\Theta}_1$ and $\mathbf{\Theta}_2$, respectively. Figure 1 shows a typical path diagram for a multilevel factor-analytic model with four items loading on one latent construct. Note that neither the factor loadings nor the number of factors needs to be the same across the within-cluster and between-cluster models.

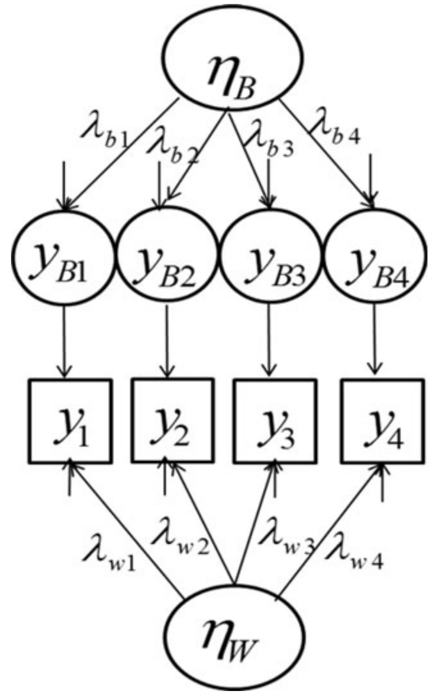

FIGURE 1 Path diagram of a multilevel factor model with four items loading on one latent construct.

MULTILEVEL DYNAMIC FACTOR MODEL (MDFM)

Multilevel factor models can be eligibly applied to multiple multivariate time series data, where serial observations (Level 1 or within-person) are considered nested within individuals (Level 2 or between-person). Within this context, the Level 1 model is indeed an application of standard common factor model to time series data, recognized as the P-technique factor analysis (Cattell, 1966) that models the within-person variability and its sources. The Level 2 model is fitted using the intraindividual means of all individuals, thereby modeling between-person variability and its sources.

Disentangling the within- and between-person variability is a great advantage of applying multilevel factor models to time series data. As shown in research, multilevel factor models help examine state and trait variations for personality-related characteristics, where traits are defined as within-person means over time and states refer to fluctuations around within-person means (Hamaker et al., 2007; Roesch et al., 2010; Timmerman, Ceulemans, Lichtwarck-Aschoff, & Vansteelandt, 2009). However, this application is subjective to the limitation inherited from the P-technique that time order information cannot be incorporated in the

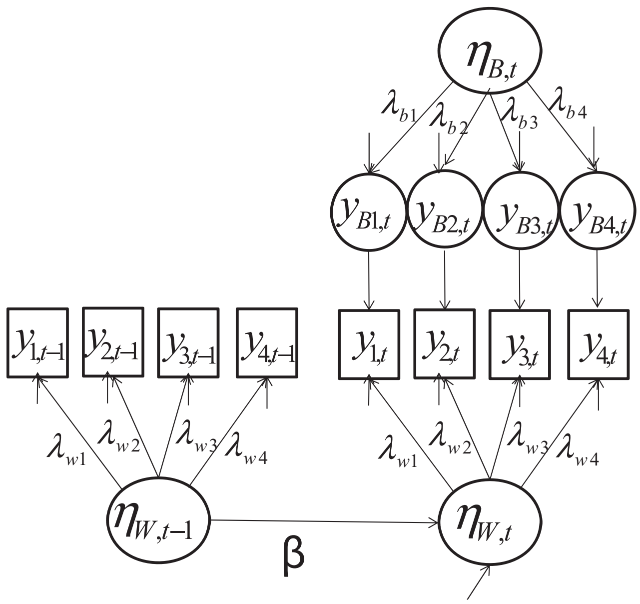

FIGURE 2 Path diagram of a lag1 multilevel dynamic factor model.

models (Jones & Nesselroade, 1990). As such, the dynamics of state characteristics cannot be modeled and thereby examined.

As an extension to multilevel factor models, MDFM allows lagged relationships existing among latent constructs in the within-person model. As indicated in Figure 2, the factor loadings at the within-person level for the same items are set to be identical between time t-1 and t, a typical assumption for dynamic factor models. The regression coefficient, β, reflects the lag1 relationship between the latent construct at t-1 and t—how the status *yesterday* influences the status *today*. At the between-person level, a common factor model is imposed on the random intercepts estimated from the within-person model. No lagged relations can be modeled at this level because intraindividual means cannot have lagged values. The specific mathematical expressions of a lag1 MDFM are stated as follows:

within-person

$$\mathbf{y}_{1,i,t} = \mathbf{\Lambda}_{1,i}\boldsymbol{\eta}_{1,i,t} + \boldsymbol{\varepsilon}_{1,i,t}, \tag{6}$$

$$\boldsymbol{\eta}_{1,i,t} = \mathrm{B}_{1,i}\boldsymbol{\eta}_{1,i,t-1} + \boldsymbol{\xi}_{1,i,t}, \tag{7}$$

between-person

$$\mathbf{y}_{2,i} = \boldsymbol{\nu}_2 + \mathbf{\Lambda}_2\boldsymbol{\eta}_{2,i} + \boldsymbol{\varepsilon}_{2,i}, \tag{8}$$

where the subscript i indices individuals and t indices measurement occasions; $\boldsymbol{\eta}_{1,i,t-1}$represents a vector of latent status at time t – 1, that is, the lag1 factor score vector; $\mathrm{B}_{1,i}$ contains autoregressive and cross-regressive coefficients between $\boldsymbol{\eta}_{1,i,t}$ and $\boldsymbol{\eta}_{1,i,t-1}$; $\boldsymbol{\xi}_{1,i,t}$ contains residual terms, normally distributed with zero means and covariance matrix $\boldsymbol{\Psi}$; all other terms share the same meanings as those defined earlier in Equations 4 and 5. Note that multilevel factor model described earlier is a special case of MDFM, when no lagged relationship is modeled, namely, Equation 7 is not included in the model. Possible extensions can be reached by adding more lagged relations into the model. For example, the influences of lag2 status ($\boldsymbol{\eta}_{1,i,t\text{-}2}$) on lag1 ($\boldsymbol{\eta}_{1,i,t\text{-}1}$) and lag0 ($\boldsymbol{\eta}_{1,i,t}$) status can be examined. As with the standard single-level DFM (i.e., Equations 1 and 2), the MDFM model is subject to the assumption that the multivariate time series are stationary over time.

It needs to be noted that the MDFM presented earlier resembles the integrated state-trait (ITS) model proposed by Hamaker et al. (2007, Table 1). Both of the models recognize the importance of analyzing within-person variability as well as the between-person variability for multiple multivariate time series data. The major difference lies in the way the models are formulated and estimated. The ITS models are estimated in two independent steps—a DFM for the within-person data and a common factor model for the between-person data (i.e., the estimated intraindividual means), whereas MDFMs are estimated simultaneously using multilevel SEM techniques. More detailed discussions are included in a later section of this article.

In typical MDFMs, the regression coefficients between latent constructs at t-1 and t are assumed to be fixed. This assumption may be too strict due to the fact that dynamics may be quite different across individuals (Ferrer & Widaman, 2008; Song & Ferrer, 2012). To relax this assumption, one may allow these coefficients in MDFMs to vary randomly in the population and possible covariates can therefore be included to explain between-person differences in these coefficients. However, the use of high-dimensional integration for model estimation in such contexts can be very time consuming and memory demanding; therefore, increasing the complexity of the model will increase the difficulty in model estimation and certain models may not be estimated. The possibility of allowing dynamical coefficients to be random is explored in this study.

EMPIRICAL ILLUSTRATION OF MDFMS

Methods

Participants

The participants in this illustration were college students from a large-scale longitudinal study that was designed to investigate emotion/affect interactions within couples in romantic relationships. As part of the study, the couples were asked to separately assess their emotional experiences specific to their relationship every day before bedtime. The couples varied greatly in their length of daily ratings. The current analysis included daily reports from couples who had at least 50 consecutive days of observations. We ended up with 205 heterosexual couples in the sample and the average number of observations was 68 days per person.

More details of the sample and procedures are available elsewhere (e.g., Ferrer & Widaman, 2008; Ferrer & Zhang, 2009; Song & Ferrer, 2009).

Measures

The questionnaire used for daily report of affects contained a set of 18 item descriptors. All items were rated using a 5-point Likert scale (1 = *very slightly or not at all;* 5 = *extremely*). Half of the items were used to tap positive affect (PA) and the other half were used to tap negative affect (NA). Previous studies have confirmed the two-factor structure with 9 items loaded on each PA and NA (Ferrer & Widaman, 2008; Song & Ferrer, 2009). For the illustrative purpose in this study, we used 4 items indicative of PA, including *emotionally intimate, trust, loved,* and *happy,* and 4 items indicative of NA, including *sad, blue, discouraged,* and *angry,* for each person in the dyads.

Analysis

Altogether, our analysis included 16 observed time series (i.e., 4 PA series and 4 NA series for the female and male in each dyad) and 4 latent variables (i.e., PA and NA the female and male in each dyad). We first ran a confirmatory multilevel factor-analytic model by which we examined the intraclass correlations (ICCs) to disentangle within and between variability for each item as well as for the latent constructs of PA and NA. Second, we ran lag0, lag1, and lag2 MDFMs with the focus on estimating and interpreting the autoregressive effects of PA and NA within person and cross-regressive effects of PA and NA between persons within dyads. Third, we demonstrated the utility of random coefficients MDFMs to multiple multivariate time series. All of the analyses were conducted by using Mplus 6.11. The full-information maximum likelihood via the Expectation-Maximization (EM) algorithm was used to handle missing data and unbalanced cluster sizes. The Mplus code for fitting the lag1 MDFM can be found in the Appendix.

Results

Multilevel Factor Analysis (i.e., lag0 MDFM)

The lag0 MDFM was specified by Equations 6 and 8. Estimating this model yielded a good fit, $\chi^2 = 2341$, $df = 196$, $p < .001$, CFI = .941, TLI = .928, RMSEA = .028, SRMR = .033 for within and .051 for between. The respective ICCs for *sad, blue, discouraged,* and *angry* were .204, .214, .217, and .173 for females and .205, .229, .222, and .236 for males. The respective ICCs for *emotionally intimate, trust, loved,* and *happy* were .379, .530, .467, and .450 for females and .413, .547, .454, and .440 for males. All the ICCs suggested a substantial amount of within- and between-individual variance. However, NA items exhibited less between-person variation than within-person variation (ICCs < .5), indicating that state variation is greater than trait variation for all NA items. PA items exhibited approximately equal amounts of within- and between-person variation (ICCs ≈ .5), evidencing equal state and trait variations.

TABLE 1
Estimates of Factor Loadings From Fitting the Multilevel Factor Model With Standardized Loadings in Parentheses

	Females	Males
Within-person		
Intimate	1.000(.617)	1.000(.581)
Trust	.798(.638)	.815(.596)
Loved	1.059(.819)	1.087(.763)
Happy	1.187(.845)	1.243(.814)
Sad	1.000(.795)	1.000(.753)
Blue	.924(.780)	.953(.746)
Discouraged	.870(.731)	.858(.663)
Angry	.783(.688)	.745(.611)
Between-person		
Intimate	1.000(.820)	1.000(.853)
Trust	1.082(.847)	1.008(.821)
Loved	1.126(.967)	1.035(.976)
Happy	1.165(.953)	1.068(.967)
Sad	1.000(.915)	1.000(.924)
Blue	.993(.936)	1.057(.948)
Discouraged	.927(.862)	1.004(.906)
Angry	.758(.846)	.894(.819)

The estimates of factor loadings from the model are shown in Table 1. As expected, the factor loadings were all statistically and practically significant. Note that when exploratory multilevel factor analysis is preferred, models with varying number of factors could be specified at each level, and statistical tests could be done to determine the best-fitting model. Neither the number of factors nor the nature of the factors needs to be same across levels.

The interfactor correlations are presented in Table 2. At the within level, females PA and NA were significantly correlated with their partners' respective affect (i.e., $r_{\text{Fp with Mp}} = .398$, $r_{\text{Fn with Mn}} = .413$, $ps < .001$), indicating that in general,

TABLE 2
Interfactor Correlations Among PA and NA From Males and Females in the Same Dyads

Within-person	2	3	4
1. Fp	.398	−.662	−.290
2. Mp		−.311	−.635
3. Fn			.413
4. Mn			
Between-person			
1. Fp	.690	−.461	−.215
2. Mp		−.257	−.333
3. Fn			.538
4. Mn			

Note. PA = positive affect; NA = negative affect. Prefix F indicates females and prefix M indicates males; all correlation coefficients are significant, $ps < .001$.

the two persons in a dyad showed good synchronization in their daily affect experiences—one feels positive (negative) today and the other tends to feel positive (negative) today as well. At the between level, these two correlations were also significant (i.e., $r_{\text{Fp with Mp}} = .690$, $r_{\text{Fn with Mn}} = .538$, $ps < .001$). It suggested that compared with people in other dyads, the two persons in same dyad exhibit similar trait levels of PA and NA. It is also noticeable that these two correlations were greater at the between level than the within level, indicating that people in dyads showed a higher degree of synchronization in their trait levels of PA and NA than in their state PA and NA. Table 2 shows that all other correlation coefficients were stronger at the within level than the between level. For example, females' PA and NA were correlated at –.662 at the within level but –.461 at the between level. The same pattern was found for males' PA and NA, –.635 at the within level but –.333 at the between level. The results suggested that relational-specific PA and NA evidence higher dependency at state level than trait level. In other words, state PA is predictive of state NA; however, trait PA is less predictive of trait NA.

ICC was also calculated manually for each factor as the ratio of between-individual variability to total variability. The ICCs suggested a slightly larger amount of trait variability than state variability for PA, $\text{ICC}_{\text{Fp}} = .520$ and $\text{ICC}_{\text{Mp}} = .600$; however, there was a much lower amount of trait variability than state variability for NA, $\text{ICC}_{\text{Fn}} = .253$ and $\text{ICC}_{\text{Mn}} = .279$. Thus, compared with PA, NA was more a state-like construct, more subjective to daily fluctuations.

Multilevel Dynamic Factor Model

Lag1 MDFM. The lag1 MDFM was specified by Equations 6, 7, and 8. The model estimation yielded $\chi^2 = 24727$, $df = 922$, $p < .001$, CFI = .873, TLI = .864, RMSEA = .043, SRMR = .027 for within and .439 for between. The factor loadings at the within and between levels were all similar to those produced in the multilevel factor-analytic model. Thus they were not reported again here to save space. The results at the within level are shown in Table 3 and in Figure 3.

TABLE 3
Estimates of Autoregressive and Cross-Regressive Coefficients Among Latent Constructs From Fitting the Lag1 Multilevel Dynamic Factor Model (MDFM)

		DVs			
		Fp	Fn	Mp	Mn
IVs	Fp_1	.467***	−.014	.056*	−.034*
	Fn_1	.082**	.340***	.020	−.001
	Mp_1	.007	−.043	.471***	.016
	Mn_1	−.037	−.006	.105***	.309***

Note. Fp = female positive affect; Fn = female negative affect; Mp = male positive affect; Mn = male negative affect; _1 = lag 1.
*$p < .05$. **$p < .01$. ***$p < .001$.

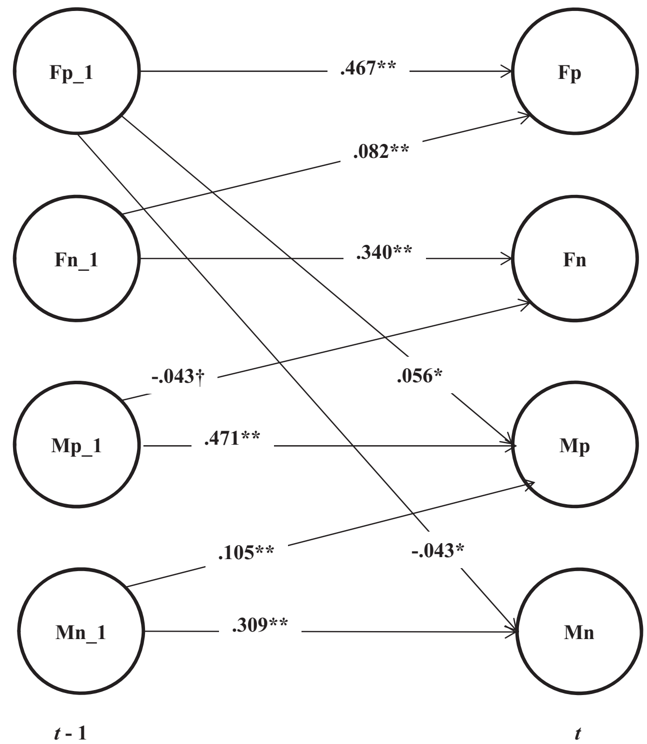

FIGURE 3 Within-person part of the lag1 MDFM to empirical data with significant parameter estimates presented only († indicates $p < .10$).

To simplify the presentation, only the significant estimates of regression coefficients are displayed. The results showed that autoregressive effects were evident for both PA and NA and for both males and females, $\beta_{\text{Mp}} = .471$, $\beta_{\text{Fp}} = .467$, $\beta_{\text{Mn}} = .309$, $\beta_{\text{Fn}} = .340$, $ps < .001$, indicating that the dyads tend to carry over their feelings about their relationships at least to the next day with PA showing stronger carry-over effect. Regarding the cross-affect effect within individual, the regression from NA at t-1 to PA at t was the only one significant, for both males ($\beta = .105$, $p < .01$) and females ($\beta = .082$, $p < .01$). It suggested that holding other influences constant, NA *yesterday* contributed positively to PA *today*; however, the opposite direction did not hold. Finally, we examined the lagged influences (t-1 to t) between males and females. It appeared that only PA at t-1 exerted influences on the other person's affect at t—female's PA contributed positively to male's PA ($\beta = .056$, $p < .05$) and negatively to male's NA ($\beta = –.034$, $p < .05$) and male's PA contributed negatively to female's NA ($\beta = –.043$, $p = .067$).

Lag2 MDFM. Autoregressive and cross-regressive relationships were specified among the latent constructs between t-2 and t. Meanwhile, we specified the regression coefficients from t-1 to t were identical to those from t-2 to t-1, a typical constraint under the assumption of stationarity. The model estimation yielded $\chi^2 = 56447.805$, $df = 2166$, $p < .001$, CFI = .813, TLI = .805, RMSEA = .042, SRMR = .066

TABLE 4
Estimates of Autoregressive and Cross-Regressive Coefficients Among Latent Constructs From Fitting the Lag2 Multilevel Dynamic Factor Model (MDFM)

		DVs			
		Fp	Fn	Mp	Mn
	Fp_1	.412***	−.098***	.100***	−.041*
IVs	Fp_2	.092**	.079***	−.060**	.011
	Fn_1	.016	.279***	.042	.014
	Fn_2	.035	.108***	−.027	−.010
	Mp_1	.074*	−.066*	.406***	−.075**
	Mp_2	−.057*	.024	.158***	.057*
	Mn_1	−.010	.022	.021	.251***
	Mn_2	−.008	−.030	.099***	.077**

Note. Fp = female positive affect; Fn = female negative affect; Mp = male positive affect; Mn = male negative affect; _1 = lag 1; _2 = lag 2.
*$p < .05$. **$p < .01$. ***$p < .001$.

for within and .484 for between. As shown in Table 4, both PA and NA had significant impacts on themselves in *next day* and *the day after next,* $\beta_{\text{Fp on Fp\_1}} = .412$, $\beta_{\text{Fp on Fp\_2}} = .092$, $\beta_{\text{Fn on Fn\_1}} = .279$, $\beta_{\text{Fn on Fn\_2}} = .108$, $\beta_{\text{Mp on Mp\_1}} = .406$, $\beta_{\text{Mp on Mp\_2}} = .158$, $\beta_{\text{Mn on Mn\_1}} = .251$, $\beta_{\text{Mn on Mn\_2}} = .077$, ps $< .01$, with PA showing stronger lag1 autoregressive relationships than NA. This analysis also showed interesting results with regard to coregulation on affects between females and males in the same dyads. Specifically, one person's PA was influenced by the other's PA *yesterday* and *the day before yesterday* but not by the other's NA, $\beta_{\text{Fp on Mp\_1}} = .074$, $\beta_{\text{Fp on Mp\_2}} = –.057$, $\beta_{\text{Mp on Fp\_1}} = .100$, $\beta_{\text{Mp on Fp\_2}} = –.060$, ps $< .05$; one person's NA was mostly influenced by his or her own PA *yesterday* and *the day before yesterday*, $\beta_{\text{Fn on Fp\_1}} = –.098$, $\beta_{\text{Fn on Fp\_2}} = .079$, $\beta_{\text{Mn on Mp\_1}} = –.075$, $\beta_{\text{Mn on Mp\_2}} = .057$, ps $< .05$, and by his or her partner's PA yesterday, $\beta_{\text{Fn on Mp\_1}} = –.066$, $\beta_{\text{Mn on Fp\_1}} = –.041$, ps $< .05$. In summary, it seems that PA was the more influential affect—it had lingering impact not only on a person's own NA but also on his or her partner's PA.

Multilevel Dynamic Factor Model With Random Dynamical Coefficients

We fitted the lag1 MDFM by allowing all autoregressive and cross-regressive coefficients to randomly vary, but the model cannot be estimated due to the memory shortage of the computer (x64_based PC with Intel (R) Core (TM)2 Duo CPU, 3.16GHz and 4.00GB memory). Then we boiled down the model to include only one affect (either PA or NA) for both females and males in dyads. These models still cannot be estimated for the same reason. In order to show the utility of MDFM with random dynamical coefficients, we finally estimated a lag1 MDFM with only PA for females while allowing the autoregressive coefficients to be random.

The estimates of raw factor loadings at the within-person level were all significant and comparable in magnitude with those obtained in previous models. However, the factor loading of *happy* at the between level was not significant ($\lambda = .298, p = .587$). Of interest in this analysis were the mean and variance of the autoregressive coefficient (β). They were estimated as .617 ($p < .001$) and .062 ($p < .001$), respectively, indicating that in general, females do carry over their positive affect to the next day and the carry-over effect varies—some females show strong predictability whereas others may lack predictability in their positive affect within 2 days of period. Potential covariates could be further added to explain such between-person variations.

Discussion

Multivariate time series data offer an opportunity for researchers to study dynamics of various systems in social and behavioral sciences. When multivariate time series data are collected from multiple units, how to synchronize dynamical information becomes a silent issue. To address this issue, the current study presented a statistical model, MDFM, which analyzes multiple multivariate time series in multilevel SEM frameworks. We applied lag0, lag1, and lag2 MDFMs to empirical data on affect from dating couples to illustrate the uses of MDFMs. We also considered a model extension where the dynamical coefficients (i.e., autoregressive and cross-regressive coefficients) were allowed to be randomly varying in the population. In the following, results are summarized and discussed with regard to the empirical application and related methodological issues.

As for the empirical application, our analysis yielded a few interesting findings. First, dependency exists between two persons in the same dyads in trait affects (PA and NA) as well as state affects, with higher dependency showing up in trait affects. Second, dependency exists between PA and NA within person at both trait and state levels, with higher dependency between state PA and NA. Third, PA shows an approximately equal amount of state and trait variations, whereas NA shows larger state variation than trait variation. Fourth, both males and females carry over their affects to the next 2 days at least. Fifth, people in romantic relationships regulate not only their own affects but also their partner's affects over time. In summary, this application successfully demonstrated the utility of MDFMs in analyzing multiple multivariate time series.

In our empirical example, our focus was on disentangling within and between sources of variability and more important, on modeling the time order effect of the latent affects. The factor structure of affects was not of much interest, as it has been well supported by previous studies (e.g., Ferrer & Widaman, 2008; Song & Ferrer, 2009, 2012). Therefore, we adopted confirmatory factor-analytic models at both the within and between levels. However, note that when there is a lack of knowledge about factor structures at either or both levels, exploratory MDFMs should be used by which both number and nature of factors could be different across within and between levels (e.g., Roesch et al., 2010; Timmerman et al., 2009).

The key idea of MDFM is to disentangle the sources of state and trait (within- and between-person) variations of the observed data as well as to model the dynamics of state aspects of human characteristics over time. This idea also underlies the integrated trait-state model (ITS model; Hamaker et al., 2007). The key difference between the two models is that MDFM is a multilevel SEM technique that simultaneously models the within and between structure of variability, whereas the ITS model estimates them as two separate and independent models, using a dynamic factor model for the within structure of variability and dynamics and a common factor model for the between structure of variability. Moreover, before applying the within-person model, the ITS model attempts to identify subgroups of homogeneous individuals with identical factor structure at within level. Although this approach has the advantage of testing for homogeneity of factorial structure, it often yields empirically many subgroups with too few individuals in each subgroup as is, for instance, illustrated by the example presented by Hamaker et al. (2007). The MDFM presented earlier assumes that factorial structure is homogeneous across individuals, which may not be realistic in some cases. A future attempt could allow factorial structure, such as factor loadings at least, to vary across individuals in MDFMs.

Under the framework of MSEM, the MDFMs demonstrated through the empirical data were estimated using time delay embedding of the observed time series; that is, the lagged series of observations are first created for each variable and then entered the model as separate, new variables to model dynamical relationships among latent series. This is different from other alternative DFM analyses, where the DFMs could be estimated by fitting the block-Toplitz matrix of variances and covariances of the observed time series (e.g., Molenaar & Nesselroade, 1998; Wood & Brown, 1994; Zhang, Hamaker, & Nesselroade, 2008), by updating iteratively the factor score series by Kalman filter or smoother using state-space modeling techniques (e.g., Chow et al., 2007; Song & Ferrer, 2009), or by maximizing raw likelihood function to obtain exact MLEs (Voelkle, Oud, von Oertzen, & Lindenberger, 2012). Due to the time delay embedding, maximum likelihood estimation produces pseudo-MLEs instead of exact MLEs. The time series literature has shown that the pseudo-ML estimates based on time delay embedding are not different from the exact ML estimates (e.g., Hamilton, 1994) under stationarity. Moreover, it was found in another study that three methods, time delay embedding, block-Toplitz matrix, and state-space model, did not produce different parameter estimates using simulated data under strict stationary conditions (Zhang & Song, 2013). However, little has been known about how well the pseudo-ML estimation performs under other less studied conditions, such as nonstationarity and multilevel data structure, with regard to the behaviors of parameter estimates, standard errors, model fit indices, and significance tests. Future research is needed to address this issue more thoroughly.

We extended the MDFMs by allowing the dynamical coefficients to randomly vary instead of being fixed as in standard MDFMs. This extension offers opportunities to examine the effect of potential covariates in explaining interindividual differences in dynamics. However, a few issues are noticeable and need to be addressed in future research. First, due to the use of high-dimensional integration for model estimation, random coefficient MDFMs are very time consuming and computer memory demanding. As a result, complexity of MDFMs could be a problem causing the models to be nonestimable. Second, the distribution of random coefficients is assumed to be normal by default in Mplus, which may not be appropriate for some cases. In our empirical example, the distribution of the autoregressive coefficients of Fp_t on Fp_{t-1} is very likely to be truncated normal, with all values being positive and bounded by one. However, overriding the default distribution cannot be done with the current version of Mplus. To solve this problem, Bayesian estimation with MCMC would be appealing to apply in future research, as pointed out by Muthén and Asparouhov (2012) that Bayesian estimation is motivated to be used when "analyses can be made less computationally demanding" and "new types of models can be analyzed" (p. 5). Indeed, a few studies have demonstrated the utility of Bayesian methods for estimating complex dynamic factor models (e.g., Chow, Tang, Yuan, Song, & Zhu, 2011; Song & Ferrer, 2012; Zhang, Hamaker, & Nesselroade, 2007).

The MDFM proposed in this article serves a general goal of integrating dynamical information from multiple multivariate time series. Its estimation was implemented by using multilevel structural equation modeling techniques. Although this application is novel, multilevel modeling of dynamics with other models exists in the literature. For example, differential structural equation models have been used to examine between-person variability in the regression coefficients relating derivatives (Boker & Ghisletta, 2001; Butner, Amazeen, & Mulvey, 2005). Moreover, researchers have shown that two contemporary cyclic state-space models (i.e., dynamic harmonic regression model and stochastic cycle model) can be extended to the multiple-subject cases, and variability on certain model parameters could be examined across individuals (i.e., Chow, Hamaker, Fujita, Boker, 2009). The MDFM proposed in this article is different from these two techniques in part in that it incorporates measurement models at the within-person level; therefore, measurement errors can be explicitly taken into account. In general, MDFM is well suited to modeling within-person dynamics and between-person variability in intraindividual means when one or more common factors are assumed to underlie the multiple time series of interest.

A couple of limitations need to be mentioned regarding MDFM and its applications. First, as pointed out earlier, the MDFM produces pseudo-ML estimates based on time delay embedding. Therefore, the MDFMs fitted to same data with different lags are not nested models anymore and thus, conventional methods for model comparison, for example, chi-square difference test, may not applicable. We recommend that each model be evaluated independently and if a model satisfies the widely accepted criteria, it could be considered a good model. Second, one of the important assumptions of MDFMs is that multivariate time series need to be stationary and normally distributed (i.e., ergodicity; Molenaar, 2004) for appropriate application and interpretation of MDFMs. For nonstationary time series, certain statistical procedures, such as differencing or detrending, can be performed to stationarilize the series. In our application, stationarity was implicitly assumed. In fact, a small number of nonstationary univariate series were identified in the previous studies where a similar data set was investigated (about 8%; see Song & Ferrer, 2012). Although the actual effect of fitting a small portion of nonstationary series using MDFM is unknown, readers should be cautious when generalizing our empirical results. Third, it is very possible that the participants in this study varied greatly in terms of their actual time points of assessment, although they were instructed to rate their affect every day before bedtime and return their daily ratings every other week to the lab. In general, the varying time intervals within and/or between individuals could be a serious issue in estimating parameters using discrete time models (e.g., Voelkle & Oud, 2013). In future research, one could either improve study designs to ensure individuals having relatively equal intervals of assessments or develop alternative statistical models to accommodate the varying time intervals.

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APPENDIX

```
DATA: file is lagDyad50.dat;
VARIABLE: names are dyad_id days
    Fsad LFsad Femint LFemint Ftrust LFtrust Fblue
      LFblue Floved
    LFloved Fhappy LFhappy Fdisc LFdisc Fangry
      LFangry Msad LMsad
    Memint LMemint Mtrust LMtrust Mblue LMblue
      Mloved LMloved Mhappy
    LMhappy Mdisc LMdisc Mangry LMangry;
      usevariable are dyad_id
    Fsad LFsad Femint LFemint Ftrust LFtrust Fblue
      LFblue Floved
    LFloved Fhappy LFhappy Fdisc LFdisc Fangry
      LFangry Msad LMsad
    Memint LMemint Mtrust LMtrust Mblue LMblue
      Mloved LMloved Mhappy
    LMhappy Mdisc LMdisc Mangry LMangry;
      cluster = dyad_id;
      missing = .;
ANALYSIS: type = general twolevel;
    ESTIMATOR = MLR;
    ITERATIONS = 10000;
    H1ITERATIONS = 10000;
MODEL:
    %within%
    Fp by Femint@1 Ftrust*(a1) Floved*(a2) Fhappy*(a3);
    Mp by Memint@1 Mtrust*(b1) Mloved*(b2)
      Mhappy*(b3);
    Fn by Fsad@1 Fblue*(c1) Fdisc*(c2) Fangry*(c3);
    Mn by Msad@1 Mblue*(d1) Mdisc*(d2) Mangry*(d3);
    Fp_1 by LFemint@1 LFtrust*(a1) LFloved*(a2)
      LFhappy*(a3);
    Mp_1 by LMemint@1 LMtrust*(b1) LMloved*(b2)
      LMhappy*(b3);
    Fn_1 by LFsad@1 LFblue*(c1) LFdisc*(c2)
      LFangry*(c3);
    Mn_1 by LMsad@1 LMblue*(d1) LMdisc*(d2)
      LMangry*(d3);

    Fp ON Fp_1; Fn ON Fn_1; Fn ON Fp_1; Fp ON Fn_1;
    Mp ON Mp_1; Mn ON Mn_1; Mn ON Mp_1; Mp ON
      Mn_1;
```

```
Mp ON Fp_1; Mp ON Fn_1; Mn ON Fp_1; Mn ON Fn_1;
Fp ON Mp_1; Fp ON Mn_1; Fn ON Mp_1; Fn ON Mn_1;

%between%
Fp_b by Femint@1 Ftrust* Floved* Fhappy;
Mp_b by Memint@1 Mtrust* Mloved* Mhappy;
Fn_b by Fsad@1 Fblue* Fdisc* Fangry*;
Mn_b by Msad@1 Mblue* Mdisc* Mangry*;

Fp_b WITH Mp_b;
Fn_b WITH Mn_b;
```